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Experiment No: 5

Date:

Experiment Title: Design of ATM System using UML Diagrams

Aim:

To create and model an ATM (Automated Teller Machine) system using UML diagrams, namely the Class Diagram and Sequence Diagram, in order to depict the structural components and operational flow of an ATM withdrawal transaction.

Problem Description:

An ATM system provides banking services to customers such as cash withdrawal, deposit, balance enquiry, and receipt generation without requiring them to visit a bank branch. The system must securely authenticate the customer by validating the card and verifying the PIN, check the account balance, process the requested transaction, and update the bank database accordingly.

The objective is to design a well-organized system model that clearly illustrates the different components of the ATM system and shows the interaction among the user, ATM machine, and bank server during a transaction process.

Given Conditions / Constraints:

- The system should authenticate the customer through card validation and PIN confirmation.
- The ATM must interact with the bank server to obtain transaction approval.
- The available account balance must be verified before performing a withdrawal.
- A receipt must be generated after successful completion of the transaction.
- The card should be returned to the user at the end of the transaction.
- The UML representation must include both a Class Diagram and a Sequence Diagram.

Approach / Methodology:

- Identify the main participants involved: User, ATM Machine, and Bank Server.
- Determine the important classes such as User, ATM, Bank, Account, and Transaction.
- Define the attributes and functions (methods) of each class.
- Establish relationships like associations and inheritance between the classes.
- Construct the Class Diagram to represent the system structure.
- Prepare the Sequence Diagram to illustrate the step-by-step transaction flow.

Planning and Estimation:

- Study and analyze the system requirements.

- Identify all necessary components of the ATM system.
- Develop the Class Diagram.
- Design the Sequence Diagram for the withdrawal operation.
- Validate and verify the correctness of the diagrams.

Scheduling / Design Considerations:

- Ensure secure authentication of customers.
- Maintain effective communication between the ATM and the bank server.
- Prevent withdrawal if the balance is insufficient.
- Represent the transaction process clearly in the diagrams.
- Accurately show relationships such as inheritance and association.

Technical Feasibility:

The ATM system model is technically achievable as it follows standard banking procedures and UML design principles.

Operational Feasibility:

The system is simple and convenient for customers to operate.

Economic Feasibility:

ATM systems reduce manual work in banks and are economical over time.

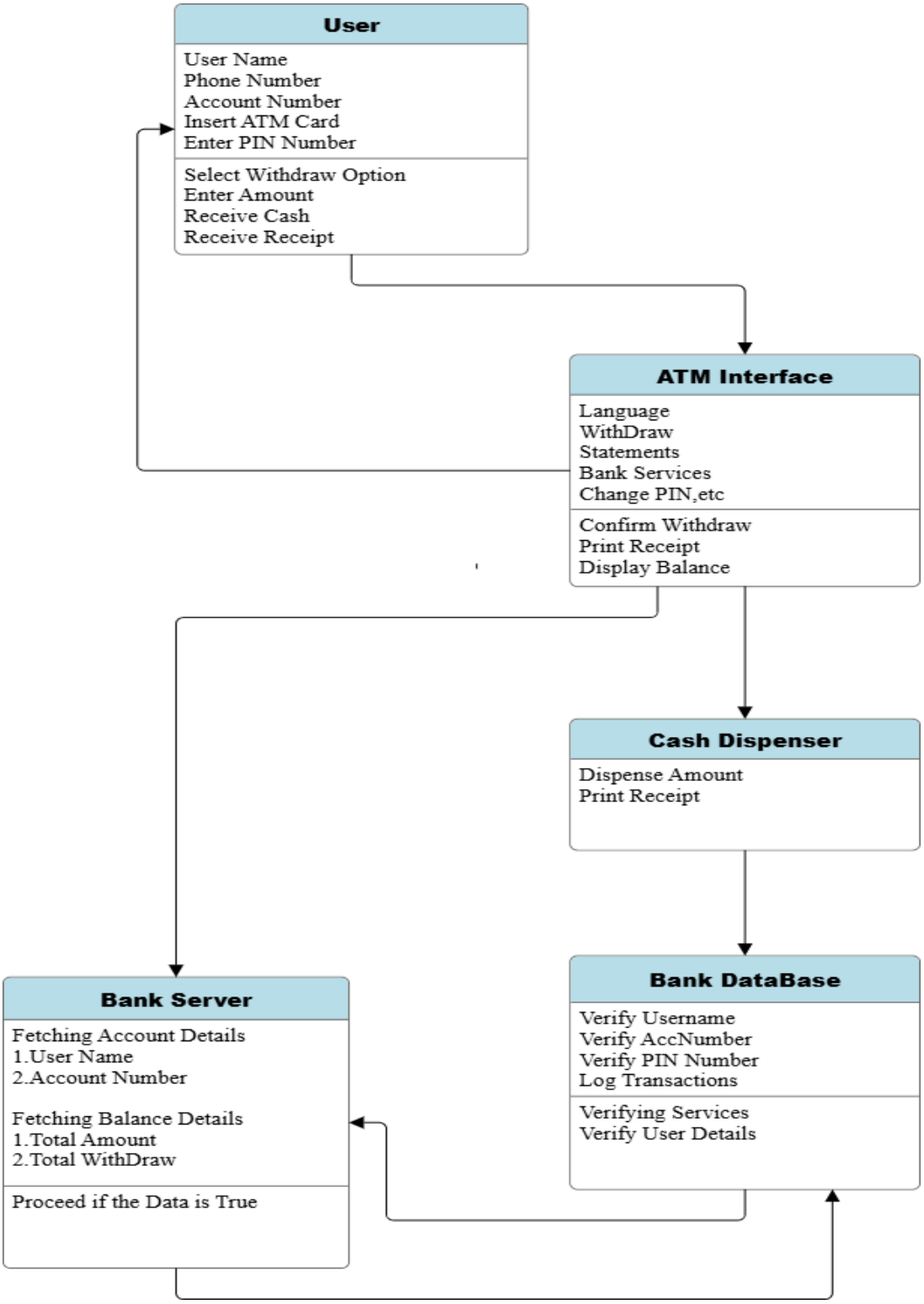
Result:

The ATM system was successfully modeled using UML Class and Sequence Diagrams. The diagrams clearly present the structural design and the sequence of operations involved in the withdrawal transaction.

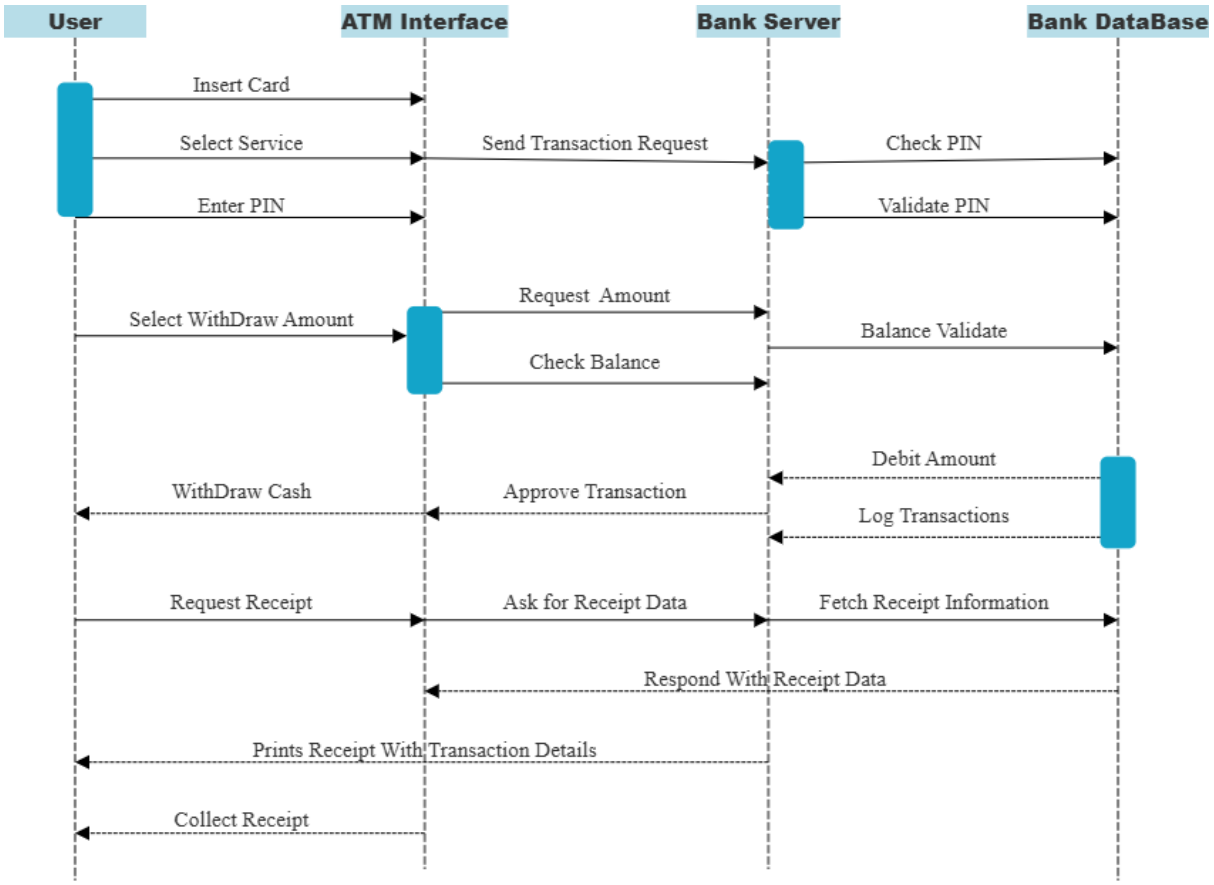
Conclusion:

The experiment effectively demonstrates the modeling of an ATM system using UML diagrams. The Class Diagram outlines the structural components and their relationships, while the Sequence Diagram explains the interaction between the user, ATM machine, and bank server during a withdrawal transaction.

UML CLASS DIAGRAM:



UML TIME SEQUENCE DIAGRAM:



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